Geometry : Vectors — problem 12.1.43

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A point P_0 and a triangle $A_1A_2A_3$ are given in a plane. Let us set $A_s = A_{s-3}$ for all $s \geq 4$. We construct the sequence $P_0, P_1, P_2, \ldots$ of points, so that the point P_{k+1} is the map of P_k rotated around A_{k+1} by 120° clockwise (mathematically negative sense) ($k = 0, 1, 2, \ldots$). Show that if $P_{1986} = P_0$, then triangle $A_1A_2A_3$ is regular (IMO 1986).